# NALOL TABLET 40mg

Each tablet contains:
Propranolol Hydrochloride 40mg

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**Product Description:** Round, red, flat, plain tablet with single score on one side only.

**Pharmacodynamics:**
Propranolol is a non selective beta-adrenergic receptor blocking agent processing no other autonomic nervous system activity. It specifically competes with beta-adrenergic receptor stimulating agents for available receptor sites. When access to beta-receptor sites is blocked by propranolol, the chronotropic, inotropic and vasodilator responses to beta-adrenergic stimulation are decreased proportionately.
The mechanism of the anti-hypertensive effect of propranolol has not been established. Among the factors that may be involved in contributing to the anti-hypertensive action are:
1. Decreased cardiac output.
2. Inhibition of renin release by the kidneys.
3. Diminution of tonic sympathetic nerve outflow from vasomotor centres in the brain.
Although total peripheral resistance may increase initially, it readjusts to or below the pre-treatment level with chronic use. Effect on plasma volume appears to be minor and somewhat variable. Propranolol has been shown to cause a small increase in serum potassium concentration when used in the treatment of hypertensive patients.
In angina pectoris, propranolol generally reduces the oxygen requirement of the heart at any given level of effort by blocking the catecholamine induced increases in the heart rate, systolic blood pressure and the velocity and extent of myocardial contraction. Propranolol may increase oxygen requirements by increasing left ventricular fibre length, end diastolic pressure and systolic ejection period. The net physiological effect of beta-adrenergic blockade is usually advantageous and is manifested during exercise by delayed onset of pain and increased work capacity. Propranolol exerts its anti-arrhythmic effects in concentration associated with beta-adrenergic blockade and this appears to be its principal anti-arrhythmic mechanism of action. In dosage greater than required for beta blockade, propranolol also exerts a quinidine-like or anaesthetic-like membrane action in the treatment of arrhythmias is uncertain.
Propranolol is almost completely absorbed from the gastrointestinal tract, but a portion is immediately bound to the liver. Peak effect occurs in one to one and a half hours. The biological half-life is approximately four hours.

**Pharmacokinetics:**
Propranolol is almost completely absorbed from the gastro-intestinal tract, but is subject to considerable hepatic tissue binding and first-pass metabolism. Peak plasma concentrations occur about 1 to 2 hours after an oral dose. Plasma concentrations vary greatly between individuals. Propranolol crosses the blood-brain barrier and the placenta. The drug is distributed into breast milk. Propranolol is more than 90% bound to plasma proteins. It is metabolised in the liver, the metabolites being excreted in the urine together with only small amounts of unchanged propranolol; at least one of its metabolites (4-hydroxypropranolol) is considered to be biologically active but the contribution of metabolites to its overall activity is uncertain. The plasma half-life of propranolol is about 3 to 6 hours. As with many other beta blockers, the biological half-life is longer than the plasma half-life. Propranolol is reported not be significantly dialyzable.

**Indication:**
a) Management of hypertension.
b) Control of cardiac arrhythmias.
c) Long term prophylaxis after recovery from acute myocardial infarction.
d) Prophylaxis of common migraine headache.
e) Management of hypertropic subaortic stenosis.
f) Management of pheochromocytoma, after primary treatment with an alpha-adrenergic blocking agent.
g) Management of angina pectoris.

**Recommended Dosage:**
The dosage range for propranolol is different for each indication.
Oral hypertension: Dosage must be individualized. The initial dosage is 40mg propranolol twice daily. Dosage may be increased gradually until adequate blood pressure is achieved. The usual maintenance dosage is 120 to 240mg per day. In some instances, a dosage of 640mg a day may be required.
Angina pectoris: Dosage must be individualized. Starting with 10-20mg three times daily, before meals and at bedtime, dosage should be gradually increased at three to seven days intervals until optimum response is obtained. Although individual patients may respond at any dosage level, the average optimum dosage appears to be 160mg per day.
Arrhythmias: 10-30mg three to four times daily before meals and at bedtime.
Myocardial infarction: The recommended daily dosage is 180-240 mg per day in divided doses, either on a three times daily or twice daily regimen. However, higher dosages may be needed to effectively treat coexisting diseases such as angina or hypertension.
Migraine: Dosage must be individualized. The initial oral dose is 80 mg propranolol daily in divided doses. The usual effective dose range is 160-240mg per day. The dosage may be increased gradually to achieve optimum migraine prophylaxis.
Hypertropic subaortic stenosis: 20-40 mg three or four times daily before meals and at bedtime.
Pheochromocytoma: Pre-operatively- 60mg daily in divided doses for three days prior to surgery, concomitantly with an alpha-adrenergic blocking agents.

**Route of Administration:** Oral

**Contraindications:**
It should not be given to patients with bronchial asthma or bronchospasm, hypoglycaemia, metabolic acidosis, sinus bradycardia or partial heart block. It is also contraindicated in patients who have exhibited hypersensitivity to it.

**Warnings and Precautions:**
Sympathetic stimulation may be a vital component supporting circulatory function in patients with congestive heart failure, and its inhibition by beta-blockade may precipitate more severe failure.
In patients without a history of heart failure, continued use of beta-blockers can, in some cases, lead to cardiac failure. Therefore at the first sign or symptom of heart failure, the patient should be digitalised and / or treated with diuretics and the response observed closely, or propranolol should be discontinued (gradually if possible).
In patients with angina pectoris, there have been reports of exacerbation of angina, and in some cases, myocardial infarction, following abrupt discontinuance of propranolol. Therefore, when discontinuance of propranolol is planned, the dosage should be gradually reduced over at least a few weeks and the patient should be cautioned against interruption or cessation of therapy without the physician advice. If propranolol therapy is interrupted and exacerbation of propranolol angina occurs, it usually is advisable to reinstitute propranolol therapy and take other measure for the management of unstable angina pectoris.
Patients with brochospastic diseases should in general not receive beta-blockers.
Propranolol should be administered with caution since it may block brochodilation produced by endogenous and exogenous stimulation of beta-receptors .The necessity or desirability of withdrawal of both beta-blocking therapies prior to major surgery is controversial. It should be noted, however, that the impaired ability of the heart to respond to reflex adrenergic stimuli may augments the risks of general anaesthesia and surgical procedures.
Beta-adrenergic blockade may prevent the appearance of certain premonitory signs and symptoms (pulse rate and pressure changes) of acute hypoglycaemia in liable insulin-dependent diabetes. In these patients, it may be more difficult to adjust the dosage of insulin.
Beta-blockade may mask certain clinical signs of hyperthyroidism. Therefore, abrupt withdrawal of propranolol may be followed by an exacerbation of symptoms of hyperthyroidism, including thyroid storm. In patients with Wolff-Parkinson-White syndrome, several cases have been reported in which after propranolol, the tachycardia was replaced by a severe bradycardia requiring a demand pace-maker.
Propranolol should be used with caution in patients with impaired hepatic or renal function.
Propranolol is not indicated for the treatment of hypertensive emergencies.
Beta-adrenoreceptors blockade can cause reduction of intraocular pressure. Patients should be told that propranolol may interfere with glaucoma screening test. Withdrawal may lead to a return of increased intraocular pressure.

**Interactions with Other Medicaments:**
The effects of other myocardial depressants agents such as quinidine, procainamide or lignocaine and drugs which interfere with calcium transport, such as verapamil, may also be enhanced by propranolol. The effects of propranolol are diminished by isoprenaline. The hypotension effects of propranolol may be reversed by adrenaline and noradrenaline. The effects of propranolol may be enhanced by guanethidine and reserpine. The hypotension effects may be enhanced by diuretics. Propranolol may enhance some of the cardiac effects of digitalis.

**Pregnancy and Lactation**
Propranolol has been shown to be embryotoxic in animal studies at doses 10 times greater than the maximum recommended human dose. As there are no adequate and well controlled studies in pregnant women, propranolol should be used during pregnancy only if the potential benefit justifies the potential risk to the foetus.
Propranolol is excreted in human milk. Caution should be exercised when propranolol is administered to a nursing women.

**Side Effects:**
The most common side effects of propranolol are nausea, vomiting, diarrhoea, fatigue and dizziness. Cardiovascular effects include bradycardia, congestive heart failure, heart block, hypotension, cold extremities and numbness of extremities. Central nervous system effects include depression, hallucination and disturbances of sleep and vision. Blood disorders and skin rashes may also occur. Other adverse effects reported include constipation, fluid retention and weight gain, muscle cramps, dry mouth and bronchospasm.

**Symptoms and Treatment of Overdose:**
Symptoms of overdosage include bradycardia, hypotension, cardiac failure, lightheadedness, weakness, fatigue, visual disturbances, nausea, vomiting, abdominal cramping, epigastric distress and bronchospasm in susceptible patients.
If ingestion is or may have been recent, evacuate gastric contents, taking care to prevent pulmonary aspiration.
Bradycardia : administer atropine (0.25 to 1mg). If there is no response to vagal blockade, administer isoprenaline cautiously.
Cardiac failure : digitalization and diuretics.
Hypotension : vasopressors e.g. levarterenol or adrenaline.
Bronchospasm: administer isoprenaline and aminophyline.

**Storage Conditions:** Store below 30°C.

**Pack Size:** Blister pack: A box of 10 x 10, 50 x 10 and 100 x 10 tablets per strip.
**Pack Size (export only):** A bottle of 1000 and 5000 tablets.

**Shelf-life:** 5 years

**FURTHER INFORMATION CONCERNING THIS DRUG CAN BE OBTAINED FROM YOUR FAMILY PHYSICIAN / LOCAL GENERAL PRACTITIONER / PHARMACIST.**

Manufacturer & Product Registration Holder:
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